Privacy attention and Behavioural response to the Cambridge Analytica Scandal

Issue attention cycles and privacy related search dynamics

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Background

- Major scandals can trigger sudden public concern, the issue-attention cycle predicts declining public concern after initial interest [1]. But less is known about the duration and nature of attention, especially in the context of privacy-related scandals.
- Online search behavior provides a real-time proxy for public interest and behavioral intent [2].
- The Cambridge Analytica scandal was a pivotal event that raised public awareness of data privacy causing different attitudes towards privacy rights[3].

Research Question :
How did public privacy concern evolve after the Cambridge Analytica scandal?

Methodology

- Privacy Concern Index = mean of relevant search terms
- Event date: 17 March 2018
- Wilcoxon signed-rank test:
  - → Nullhypothesis: Difference between the medians equals zero
  - → Alternative Hypothesis: Interest 6 months after the scandal is still higher than before inital rise
- Recovery = return to pre-event baseline (14-day mean)

Data

Google Trends (daily data)

- Feb - Oct 2018

Keywords grouped into:

- Action-oriented (e.g., delete Facebook)
- Information-oriented (e.g., Cambridge Analytica scandal)

Hypothesis

- **H1: Privacy-related searches increase immediately after the scandal.**
- **H2: After a scandal subsides, average search interest remains higher than before the event.**
- **H3: Action-oriented searches normalize faster than information-oriented searches.**

Results

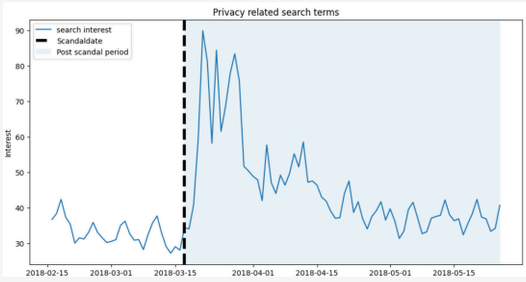

Figure 1: Sharp spike in privacy related Google searches following the Cambridge Analytica scandal

H1: Privacy-related searches increase immediately after the scandal.

Privacy related search interest doubled immediately after the scandal.

- +31.6 interest points within 14 days
- $p < .001$, very large effect ($r \approx 0.96$)

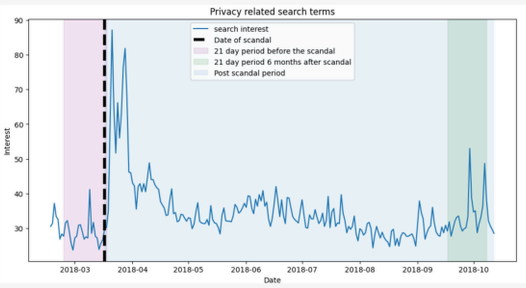

Figure 2: Analysis of a 21 day period before the scandal date and one six months later visualized

H2: Average search interest remains higher even after a scandal subsidies.

Comparison of two 21 day windows

- the first one being before the scandal
- the second one after six months have passed

→ the calculated one-sided Wilcoxon signed-rank test showed that the values were significantly lower 21 days before the scandal (Mdn= 28.61) than after 6 months (Mdn=34.01) $p = 0.0029$, alternative='less', $r = 0.71$).

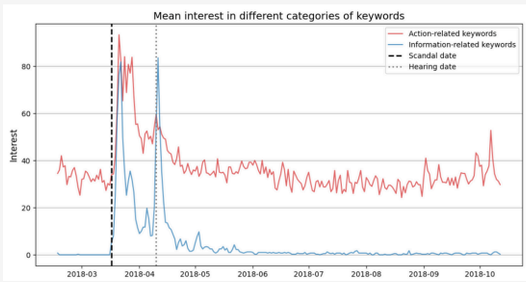

Figure 3: Comparison between action-related queries and information-related queries

H3: Action-oriented searches normalize faster than information-oriented searches.

- Action-related queries returned to baseline within 32–102 days (mean = 81), whereas information-related queries required 49–125 days (mean = 74,6).
- Some information-related queries never return to their baseline.
- Descriptive evidence that action-oriented serches normalize faster, but no statistically significant evidence.

Conclusion

- The Cambridge Analytica scandal triggered a sharp and immediate increase in privacy-related search activity.
- While overall interest declined over time, search activity remained elevated compared to pre-scandal levels, suggesting a persistent shift in baseline privacy concern.
- Action-oriented searches showed a more short-lived response, whereas information-oriented queries remained elevated for longer, indicating different temporal dynamics of behavioral intent and information seeking.
- Overall, the results support issue-attention cycle theories while highlighting that privacy scandals can have lasting effects on public awareness.

Limitations

- Google Trends data capture relative search interest rather than actual behavior and may reflect media attention as much as individual concern.
- Google Trends only allows daily data for a timespan up to nine months, which makes observations over a longer period of time rather difficult.
- Small number of keywords per category limits formal statistical inference, particularly for comparisons between action- and information-oriented queries.

Related literature

1 Downs, A.M. (1972). Up and Down with Ecology--The Issue Attention Cycle.

2 Turner, A., & Zieliński, M. W. (2021). Public Interest in Surveillance, Privacy and Data Protection. Google Big Data in Comparative Sociology. *Google Big Data in Comparative Sociology (July 24, 2023). PRZEGLĄD SOCJOLOGICZNY*, *70*(2), 129-156ISSN.

3 Frank M. Shipman and Catherine C. Marshall. 2020. Ownership, Privacy, and Control in the Wake of Cambridge Analytica: The Relationship between Attitudes and Awareness. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–12. https://doi.org/10.1145/3313831.3376662